

Hall Ticket No-

Code: 22323

Stanley College of Engineering and Technology for Women(A)

Chapel Road , Abids

B.E I Semester (Main) Semester End Examinations March-2022

**Subject :Chemistry
(IT)**

Time: 3 hours

Max.Marks:60

PART-A

Note:Answer all questions (Compulsory)

5 X2=10 M

- 1 500 ml of a sample of water contains 43.8 mg of $\text{Mg}(\text{HCO}_3)_2$, 58.5 mg of NaCl, 38 mg of MgCl_2 , 2.43 mg of $\text{Ca}(\text{HCO}_3)_2$ and 6.8 mg of CaSO_4 . Calculate its temporary, permanent and total hardness. 2 M
- 2 Under standard conditions, what happens when Zn foils are placed in AgNO_3 solution? Justify your answer. (E° of $\text{Zn}^{2+}/\text{Zn} = -0.769\text{V}$ and $\text{Ag}/\text{Ag}^+ = -0.8\text{V}$). 2 M
- 3 Outline the preparation of Buna-S, mention any two properties and applications. 2 M
- 4 Enumerate the requisites of a good fuel. 2 M
- 5 Appraise the significance of clean technology with appropriate examples. 2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Justify the use of two indicators in the estimation of alkalinity of water. In a titration, 100 ml of a water sample required 16 ml of 1/50 N HCl with phenolphthalein indicator and another 24 ml of the same HCl with methyl orange indicator. Determine the type and amount of alkalinity present in the water sample. 5 M
- b. What are scales and sludges? How are they formed in boilers? Discuss the consequences of their formation in boilers. Recommend a method to prevent them. 5 M
- OR
- c. Discuss about the principle and working of reverse osmosis. Enlist the advantages, limitations and applications. 5 M
- d. Write the principle of cathodic protection and discuss impressed current cathodic protection along with its advantages and applications. 5 M
- 7 a. Derive the equation that gives the dependence of electrode potential on concentration of electrolyte. When a metal electrode is coupled with SCE, EMF of the cell is found to be 0.49V when the $[\text{M}^{+2}] = 1\text{M}$ at 298K. EMF of the following cell, $\text{Zn(s)}/\text{Zn}^{2+}(1\text{M}) // \text{M}^{+2}(1\text{M})/\text{M(s)}$ is 0.61V. Calculate EMF of Zn-M cell when $[\text{Zn}^{+2}] = 0.01\text{M}$ at 298K. 5 M
- b. Illustrate the electrochemistry of Lithium ion cell. Write its merits and applications. 5 M
- OR
- c. Describe the construction and working of calomel electrode with a neatly labeled diagram. Write its advantages and limitations. 5 M

P.T.O

- d. Explain the construction and working of Zn-Carbon cell. Can we recharge the cell? Justify your answer. 5 M
- 8 a. Give an overview of the preparation of Bakelite from its monomers and explain whether the polymer can be recycled or not. 5 M
- b. Illustrate the molecular orbital energy level diagram of NO. 5 M
- OR
- c. With appropriate examples, discuss about the various types of polymerization. 5 M
- d. Explain the mechanism of conduction in polyacetylene. 5 M
- 9 a. A fuel has the following composition. 82% carbon, 6 % hydrogen, 1.2 % sulphur, 1.4% nitrogen and 3% ash. Calculate the gross, net calorific value and the volume of air at STP, required for the complete combustion of 2 Kg of the fuel. 5 M
- b. Explain the proximate analysis of coal and give its significance. 5 M
- OR
- c. Explain the fractional distillation of petroleum, mention the important fractions obtained in this process, along with their uses. 5 M
- d. Explain knocking, octane number and cetane number. 5 M
- 10 a. Vegetable oils cannot be used as fuel directly in IC engines. How do you overcome this problem and convert them in to biodiesel? Discuss the chemistry involved in it. 5 M
- b. Define and classify composite materials. Discuss about the constituents of composite materials. 5 M
- OR
- c. Discuss about carbon nanotubes, quantum dots and nanowires. 5 M
- d. Enumerate the applications of composite materials 5 M

Hall Ticket No-

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B.E I Semester (Main) Semester End Examinations March-2022

**Programming for Problem Solving
(Common to all Branches)**

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 What is a flowchart? How it is different from an algorithm? 2 M
- 2 Write the syntax of switch statement? 2 M
- 3 Why is it necessary to give the size of an array in an array declaration? 2 M
- 4 Write the advantages and disadvantages of using pointers. 2 M
- 5 Compare and contrast text file with binary file. 2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. What are the steps involved in creating, compiling, and executing a C program? 5 M
- b. Define flow chart. List the symbols used in flow charts. 5 M

OR

- c. Define data type? Explain about different data types with examples. 5 M
 - d. Explain about bitwise operators. 5 M
- 7 a. Differentiate between if statement and if-else statement with suitable examples and proper syntax. 5 M
 - b. Write a program to find the factorial of a given integer number using 'while' loop. 5 M

OR

- c. Write a C program to check whether the given number is prime or not. 5 M
 - d. Show how break and continue statements are used in a C program, with example 5 M
- 8 a. Demonstrate different string handling functions available in C language. 5 M
 - b. Write a C program to find the largest element in an array 5 M

OR

- c. How passing parameters to function works? Explain 5 M
- d. List and explain the different types of storage class. 5 M

P.T.O

- 9 a. Explain defining and initializing a structure with an example 5 M
- b. What is pointer? How to initialize and declare pointer variables? Explain with examples. 5 M

OR

- c. How are the members of a 'Union' are initialized and accessed? 5 M
- d. List and explain the functions used to allocate and free memory dynamically. 5 M
- 10 a. Illustrate the different file opening modes that can be used with fopen(). 5 M
- b. What are the functions used for accessing files randomly? Explain with examples. 5 M

OR

- c. Devise an algorithm for linear search and explain with an illustration. 5 M
- d. Explain selection sort operation with an example 5 M

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B.E I Semester (Main) Semester End Examinations March-2022

Mathematics-I

(Common to all Branches)

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2 = 10 M

- 1 Determine the nature of the series $\sum_{n=1}^{\infty} \frac{3n+1}{2n+7}$ 2 M
- 2 Find the equation of envelope of the family of curves $py + p^2x - 10 = 0$ where p is a parameter 2 M
- 3 If $x^y = y^x$ then find $\frac{dy}{dx}$, 2 M
- 4 Find the maximal directional derivative of x^3y^2z at P(1,-2,3) 2 M
- 5 Explain Bisection method. 2 M

PART-B

Note: Answer all questions

5 X 10 = 50M

- 6 (a) Test for convergence of the series $\sum_{n=1}^{\infty} \frac{n^3}{3^n}$ 5 M
(b) Discuss the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ 5 M
OR
(c). Test for convergence of the series $\sum_{n=1}^{\infty} \frac{n!}{(n^n)^2}$ 5 M
(d). Discuss the absolute convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{(n+1)}$ 5 M
- 7 (a). State and prove Rolle's mean value theorem. 5 M
(b) Discuss the applicability of Rolle's theorem for the function $f(x) = (x-4)^3(x-5)^2$ on [4,5] 5 M
OR
(c). Find the radius of curvature of the curve $y = x^3(x-4)$ at P(4,0) 5 M
(d). Find the Evolute of the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 5 M

8 (a). If $u = F(x - y, y - z, z - x)$, then evaluate $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ 5 M

(b). If $u = x + y + z, uv = y + z, uvw = z$ then $\frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$ 5 M

OR

(c). Find the maximum and minimum values of $x^3 + y^3 - 3axy$ 5 M

(d). Find The extreme value of xyz when $x + y + z = 18$ 5 M

9 (a). Show that $\nabla^2 r^n = n(n+1)r^{n-2}$. 5 M

(b). Show that $\nabla \times (\nabla f) = 0$, where f is a scalar point function. 5 M

OR

(c)&(d). Verify Greens theorem for $\oint_C (x + y) dx + x^2 dy$ where C is the triangle with vertices $(0,0), (2,0)$ and $(2,4)$ taken in that order 10 M

10 (a). Using Newton Raphson method find a root of the equation $x^3 - 2x - 5 = 0$ 5 M

(b). Using Gauss-Seidel method solve the system of equations $28x + 4y - z = 32, x + 3y + 10z = 24, 2x + 17y + 4z = 35$ 5 M

OR

(c)&(d). Find $f(x)$ from the following data using Newtons Divided difference interpolation 10M

x	-1	0	3	6	7
f(x)	3	-6	39	822	1611

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B.E I Semester (Main) Semester End Examinations March-2022
BASIC ELECTRICAL & ELECTRONIC CIRCUITS

(IT)

Time: 3 hours

Max Marks-60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | |
|---|---|-----|
| 1 | What are the limitations of Superposition Theorem? | 2 M |
| 2 | Can we apply KCL and KVL to AC circuits? Justify the same. | 2 M |
| 3 | Define Diffusion current and drift current of a PN junction ? | 2 M |
| 4 | What are the advantages of hybrid parameters of BJT? | 2 M |
| 5 | State "Barkhausen criteria"? | 2 M |

PART-B

5 X 10= 50M

Note: Answer all questions

- 6 a) Find the branch currents I_1, I_2, I_3 in the Fig1. Shown below using Superposition theorem .

5 M

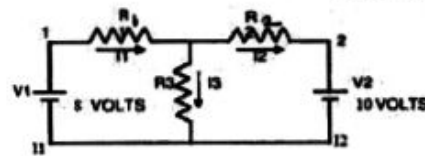


Fig1.

- b) Find the current flowing through 20Ω resistor by first finding a Norton's equivalent circuit to the left of terminals A and B.-Fig2

5 M

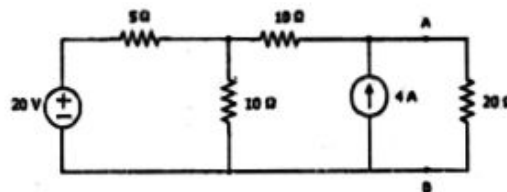


Fig2

OR

- c) Calculate the current that flows in the 1Ω resistor in the following circuit-Fig3.

5 M

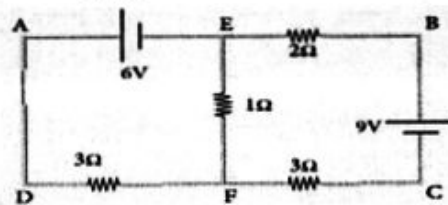
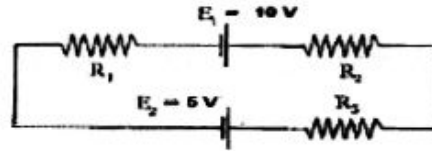


Fig3

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d) If $R_1 = 2\Omega$, $R_2 = 4\Omega$, $R_3 = 6\Omega$, determine the electric current that flows in the circuit below.



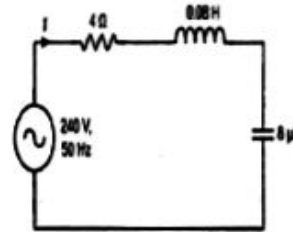
5 M

Fig4

7

a) A 240 V, 50 Hz AC supply is applied a coil of 0.08 H inductance and 4Ω resistance connected in series with a capacitor of $8\mu F$. Calculate the following –

Impedance, Circuit current, Phase angle between voltage and current, Power factor, Power consumed, Q-factor of the circuit at resonant frequency.



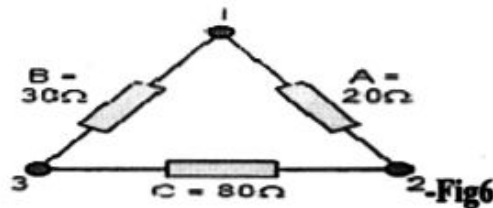
5 M

Fig5.

b) Describe various applications of RLC Circuits?

c) Differentiate between Star and Delta Connections with examples
d) Covert Delta to Star transformation using fundamentals?

OR



5 M

5 M

5 M

Fig6

8 a) Draw the block diagram of CRO and explain its operation.

5 M

b) What are the applications of CRO?

5 M

OR

c) Differentiate between Avalanche vs Zener breakdown mechanism?

5 M

d) Describe the Forward & Reverse bias characteristics of PN junction diode with neat sketches? Explain significance of Knee voltage?

5 M

9 a) Draw the h-parameter model of a BJT-CE amplifier and derive the equations for voltage gain, current gain, input impedance and output impedance.

5 M

P.T.O

b) Compare and contrast CB, CE and CC BJT configuration. - 5 M

OR

c) Draw the schematic of PNP transistor and describe the Current components in detail? - 5 M

d) Describe with neat sketch, working of n-channel JFET. - 5 M

10 a) The gain of an amplifier without feedback is 50 whereas with negative voltage feedback, it falls to 25. If due to ageing, the amplifier gain falls to 40, find the percentage reduction in stage gain (i) without feedback and (ii) with negative feedback. - 5 M

b) With a negative voltage feedback, an amplifier gives an output of 10 V with an input of 0.5 V. When feedback is removed, it requires 0.25 V input for the same output. Calculate (i) gain without feedback (ii) feedback fraction m_v . - 5 M

OR

c) Describe various applications of OP-AMP with neat sketches? - 5 M

d) Draw the schematic and working principle of RC phase shift oscillator. - 5 M